

Comment on the solutions to the Poisson equation for a spherical charge distribution

If we know the charge distribution we can obtain the potential from Coulomb's law (here in spherical coordinates):

$$V(\mathbf{r}') = \frac{1}{4\pi\epsilon_0} \int_0^\infty \int_0^\pi \int_0^{2\pi} \frac{\rho(\mathbf{r})}{|\mathbf{r} - \mathbf{r}'|} r^2 dr \sin\theta d\theta d\phi \quad (1)$$

When the charge density is spherically symmetric we may choose the orientation of the coordinate system such that $\mathbf{r}'$ lies on the z-axis, and thus the angle between the two vectors $\mathbf{r}'$ and $\mathbf{r}$ is θ . With a constant charge density between R_{min} and R_{max} This gives

$$\begin{aligned} V(\mathbf{r}') &= \frac{1}{4\pi\epsilon_0} \frac{Q}{V} \int_{R_{min}}^{R_{max}} \int_0^\pi \int_0^{2\pi} \frac{1}{\sqrt{r^2 + r'^2 - 2rr'\cos\theta}} r^2 dr \sin\theta d\theta d\phi \\ V(\mathbf{r}') &= \frac{1}{4\pi\epsilon_0} \frac{Q}{V} 2\pi \int_{R_{min}}^{R_{max}} \int_{-1}^1 \frac{1}{\sqrt{r^2 + r'^2 - 2rr'z}} r^2 dr dz, \end{aligned} \quad (2)$$

where the last equation is obtained after the variable transformation $z = \cos\theta$. This last expression can be used to obtain the potential at a any point $\mathbf{r}'$

$$V(r') = \frac{1}{4\pi\epsilon_0} \frac{Q}{V} 4\pi \int_{R_{min}}^{R_{max}} \frac{1}{r_{>}} r^2 dr \text{ where } r_{>} = r' \text{ if } r' > r, r_{>} = r \text{ if } r' < r \quad (3)$$

and we see that the potential only depends on the radial coordinate r' . To evaluate the integral for points between R_{min} and R_{max} we divide it up in tow parts: from $r = R_{min}$ to $r = r'$, and then from $r = r'$ to $r = R_{max}$. In the first region $r < r'$ and in the second $r > r'$, and we can write

$$\begin{aligned} V(r') &= \frac{1}{4\pi\epsilon_0} \frac{Q}{V} 4\pi \int_{R_{min}}^{R_{max}} \frac{1}{r_{>}} r^2 dr \\ &= \frac{1}{\epsilon_0} \frac{Q}{V} \left(\frac{1}{r'} \int_{R_{min}}^{r'} r^2 dr + \int_{r'}^{R_{max}} r dr \right) \\ &= \frac{1}{\epsilon_0} \frac{Q}{V} \left(\frac{r'^2}{3} - \frac{R_{min}^3}{3r'} + \frac{R_{max}^2}{2} - \frac{r'^2}{2} \right) \\ &= \frac{1}{\epsilon_0} \frac{Q}{V} \left(\frac{R_{max}^2}{2} - \frac{1}{3} \left(\frac{R_{min}^3}{r'} + \frac{r'^2}{2} \right) \right) \end{aligned} \quad (4)$$

This is the analytical form for the potential inside a spherical shell. If we put $R_{min} = 0$ we recover the form of the potential inside a uniformly charged sphere:

$$V(r')_{R_{min}=0} = \frac{1}{\epsilon_0} \frac{Q}{V} \left(\frac{R_{max}^2}{2} - \frac{1}{3} \frac{r'^2}{2} \right) = \frac{Q}{4\pi\epsilon_0} \frac{1}{2R_{max}} \left(3 - \left(\frac{r'}{R_{max}} \right)^2 \right) \quad (5)$$

To treat the r -dependent charge distribution from the hydrogen ground state we can use a similar approach, but the integrals get more complicated:

$$\begin{aligned} V(r')_{hyd\ 1s} &= \frac{1}{4\pi\epsilon_0} \int_0^\infty \frac{\rho(r)}{r_{>}} r^2 dr \\ &= \frac{1}{4\pi\epsilon_0} \left(\frac{1}{r'} \int_0^{r'} \rho(r) r^2 dr + \int_{r'}^\infty \rho(r) r dr \right) \\ &= \frac{e}{4\pi\epsilon_0} \left(\frac{4}{r'} \int_0^{r'} e^{-2r} r^2 dr + 4 \int_{r'}^\infty e^{-2r} r dr \right) \\ &= \frac{e}{4\pi\epsilon_0} \left(\frac{1}{r'} - e^{-2r'} \left(\frac{1}{r'} + 1 \right) \right) \end{aligned} \quad (6)$$